

Assignment 1: Project Risk, Security, and Compliance Assessment

Project Title

BantayBarangay: A Web-Based Barangay Complaint and Incident Reporting System with Blockchain-Based Document Verification

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Task 1: Identify Project Risks

BantayBarangay involves web development, database management, document handling, and blockchain-based document verification. Because of these technologies and project requirements, several risks may occur during development and implementation.

1. Lack of Blockchain Knowledge

The development team has limited experience with blockchain technology. Learning and understanding blockchain concepts, hashing, verification, and ledger implementation may take additional time and could result in mistakes during development.

2. Blockchain Implementation Issues

The blockchain-based document verification feature may not work as expected. Incorrect hashing, verification errors, or problems integrating the blockchain component with the Django application could affect the reliability of document verification.

3. Schedule Delays

The project may take longer than planned because the team needs additional time to learn blockchain technology, troubleshoot technical problems, develop system modules, and perform testing.

4. Data Privacy and Cybersecurity Threats

BantayBarangay handles resident information, complaints, incident reports, documents, and evidence. Unauthorized access, hacking, malicious input, or data breaches could expose or compromise these records.

5. Scope Changes

Additional features or changes requested during development may increase the workload. Changes to the complaint system, document management, or blockchain verification could require additional development and testing time.

Task 2: Risk Register

ID	Risk	Description	Probability	Impact	Risk Level	Response Strategy	Owner
R001	Blockchain Knowledge	development team has limited experience with blockchain technology and may need additional time to learn and understand its concepts and implementation.	High	High	High	Mitigate	Development Team
R002	Blockchain Implementation Issues	Errors in hashing, verification, or integration may cause incorrect document verification or affect system functionality.	Medium	High	High	Mitigate	Development Team
R003	Schedule Delays	Learning new technologies, debugging, testing, and system revisions may cause the project to exceed its planned schedule.	High	High	High	Mitigate	Project Manager

R004	Data Privacy and Cybersecurity Threats	Unauthorized access, hacking, malicious input, or data breaches may expose resident information and other confidential records.	Medium	High	High	Mitigate	Development Team
R005	Scope Changes	Additional requirements or requested features may increase the development workload and affect the original project plan	Medium	Medium	Medium	Avoid	Project Manager

Task 3: Identify the Most Critical Risk

Most Critical Risk: Lack of Blockchain Knowledge

The most critical risk for BantayBarangay is lack of blockchain knowledge because blockchain-based document verification is one of the distinguishing features of the proposed system. The development team needs to learn and properly implement concepts that may be unfamiliar compared with conventional web and database development.

1. Why is this risk critical to the project?

This risk is critical because insufficient knowledge of blockchain technology could lead to incorrect implementation of the document verification mechanism. The team may encounter difficulties understanding hashing, maintaining the verification records, and integrating the verification process into the Django application.

If these problems are not addressed early, they could result in repeated revisions, implementation errors, and additional testing. They may also affect the project's ability to deliver the blockchain-based document verification feature as planned.

2. What project areas will it affect?

The risk can affect several project areas:

- Schedule – More time may be required to learn and implement the technology.
- Quality – Lack of experience may result in errors in the verification process.

- Scope – The team may need to modify or simplify the blockchain feature if implementation becomes too difficult.
- Security – Incorrect implementation could weaken the reliability of document verification.

The Schedule and Quality areas are the most directly affected.

3. What response strategy will you use?

The response strategy will be Mitigate.

The team will reduce this risk by:

- Studying the blockchain concepts required for the project before full implementation.
- Creating a small prototype of the document verification mechanism.
- Testing hashing and verification separately before integrating them into the main system.
- Using reliable documentation and learning resources.
- Performing repeated testing of document verification.
- Keeping the blockchain implementation lightweight and limited to the project's actual requirements.
- Allowing sufficient development time for learning and troubleshooting.

Task 4: Security and Privacy Check

1. Does your system collect personal or sensitive data?

Yes.

BantayBarangay collects personal information from residents, including information needed for registration and identification. The system may also store complaint details, incident reports, supporting documents, and evidence.

Because these records may contain private information, access should be limited to authorized users and the data should be properly protected.

2. What possible cybersecurity threats can affect your system?

The possible cybersecurity threats include:

- Unauthorized access to resident and complaint records
- SQL injection attacks against the database
- Cross-site scripting (XSS) through malicious user input
- Brute-force attacks against user accounts
- Data breaches that expose personal information
- Malicious file uploads containing harmful content
- Unauthorized modification or deletion of records
- Server attacks that make the system unavailable
- Unauthorized modification of documents that could affect document verification

3. Which part of the CIA Triad is most important for your system?

Integrity is the most important part of the CIA Triad for BantayBarangay.

The system stores complaints, incident reports, documents, and evidence. These records need to remain accurate and should not be altered without authorization.

Integrity is particularly important because the project includes blockchain-based document verification. Hashing can be used to detect whether a document has been modified after its original verification record was created.

However, Confidentiality and Availability are also important. Confidentiality protects residents' personal information from unauthorized users, while Availability ensures that authorized residents and barangay officials can access the system when needed.

4. How will your group protect user data?

The group will implement several security measures to protect user data:

1. User Authentication – Users must provide valid login credentials before accessing protected system features.
2. Role-Based Access Control – Users will only be allowed to access functions and information appropriate to their assigned roles.
3. Password Hashing – Passwords will be stored using secure hashing instead of plain-text passwords.
4. Input Validation – User input will be validated and sanitized to reduce the risk of malicious input and attacks.
5. Database Security – Database credentials and access will be properly protected, and unnecessary direct access to the database will be restricted.
6. HTTPS – Secure communication will help protect information transmitted between users and the web application.
7. Regular Backups – Important system data will be backed up to help recover information in case of accidental deletion, system failure, or other incidents.
8. Audit Logs – Important activities, particularly actions involving complaints, documents, and records, can be logged for monitoring and accountability.
9. Blockchain-Based Document Verification – Document hashes will be used to help verify whether a document has been modified.
10. Access Control for Documents and Evidence – Documents and evidence will only be accessible to authorized users.